Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



Docker Fundamentals for Beginners

Course Code: C1285

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	3 October 2026	First version — one-day lesson plan (Docker + Kubernetes), 7.5 instructional hours.	Tertiary Infotech Academy Pte Ltd

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Course Title	Docker Fundamentals for Beginners
Course Code	C1285
Duration	1 Day (7.5 instructional hours)
Daily Schedule	9:30 AM – 5:30 PM (7.5 instructional hours, excluding lunch)
Lunch Break	12:55 PM – 1:25 PM (30 minutes)
Delivery Mode	Instructor-led, hands-on labs in KillerCoda browser terminals
Prerequisites	Basic command-line familiarity; no prior Docker or Kubernetes experience required
Tools	Docker, Docker Compose, kubectl, KillerCoda, a Docker Hub account

Course Overview

This one-day hands-on course teaches participants to containerise, run and orchestrate real applications with Docker and Kubernetes. Throughout the course learners build and deploy a single sample web application — TaskBoard (a Flask task tracker backed by Redis and PostgreSQL) — progressing from a single container to a multi-service Docker Compose stack and finally a scalable Kubernetes deployment. The morning covers Docker: commands, building images with a Dockerfile, CMD vs ENTRYPOINT, volumes, networking, configuration and Docker Hub. The afternoon covers Docker Compose and Kubernetes: Pods, Namespaces, Deployments, rolling updates and rollbacks, Services, storage, and Jobs/CronJobs.

Learning Outcomes

By the end of this course, participants will be able to:

- Explain containerisation and the Docker architecture, and manage containers with the Docker CLI.
- Build custom images with a Dockerfile, applying layer caching and .dockerignore best practices.
- Distinguish CMD from ENTRYPOINT and package command-line tools correctly.
- Persist and share data using named volumes and bind mounts.
- Connect containers over a custom network using DNS, and publish services with port mapping.
- Configure containers at runtime with environment variables and publish images to Docker Hub.
- Define and run multi-service applications (web + cache + database) with Docker Compose.
- Explain Kubernetes architecture and deploy applications as Pods, Deployments and Services.
- Scale and self-heal workloads and perform zero-downtime rolling updates and rollbacks.

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- Persist data with PersistentVolumes/Claims and run batch work with Jobs and CronJobs.

Day 1 · 9:30am – 5:30pm

Docker & Kubernetes — Build, Compose, Deploy & Scale

Morning Session · 9:30 AM – 12:55 PM

Time	Topic / Activity	Duration
9:30 – 9:40	Welcome, course objectives & overview	10 min
9:40 – 10:00	Topic 1: Containers, Docker architecture & core concepts	20 min
10:00 – 10:15	Lab 1: Docker Commands — Run & Manage Containers	15 min
10:15 – 10:30	Lab 2: Images & Inspecting Containers — pull, cp, inspect	15 min
10:30 – 10:40	Topic 2: Building images — Dockerfile, layers & build cache	10 min
10:40 – 11:00	Lab 3: Build the TaskBoard Image with a Dockerfile	20 min
11:00 – 11:15	Lab 4: Dockerfile Best Practices — .dockerignore & layer caching	15 min
11:15 – 11:35	Lab 5: CMD vs ENTRYPOINT — the taskboard-cli tool	20 min
11:35 – 11:45	Topic 3: Docker storage — volumes & bind mounts	10 min
11:45 – 12:05	Lab 6: Docker Storage — Named Volumes & Bind Mounts	20 min
12:05 – 12:15	Topic 4: Docker networking — bridges, DNS & ports	10 min
12:15 – 12:35	Lab 7: Docker Networking — Custom Bridge, DNS & Port Mapping	20 min
12:35 – 12:45	Lab 8: Configuration with Environment Variables	10 min
12:45 – 12:55	Lab 9: Sharing Images — Push to Docker Hub	10 min
Time	**Topic / Activity**	**Duration**
12:55 – 1:25	***Lunch Break***	30 min

Afternoon Session · 1:25 PM – 5:30 PM

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Time	Topic / Activity	Duration
1:25 – 1:35	Topic 5: Docker Compose — multi-service apps	10 min
1:35 – 1:45	Lab 10: Docker Compose — Single Service	10 min
1:45 – 1:55	Lab 11: Docker Compose — Multi-Service (Web + Redis)	10 min
1:55 – 2:10	Lab 12: Docker Compose — Full-Stack (Web + PostgreSQL + Redis)	15 min
2:10 – 2:35	Topic 6: Kubernetes architecture & kubectl	25 min
2:35 – 2:55	Lab 13: Kubernetes Pods — Imperative & Declarative	20 min
2:55 – 3:10	Lab 14: Kubernetes Namespaces — Environment Isolation	15 min
3:10 – 3:20	Topic 7: Deployments, ReplicaSets & self-healing	10 min
3:20 – 3:45	Lab 15: Deployments — Scaling & Self-Healing	25 min
3:45 – 4:05	Lab 16: Rolling Updates & Rollbacks	20 min
4:05 – 4:15	Topic 8: Services & networking in Kubernetes	10 min
4:15 – 4:35	Lab 17: Services — ClusterIP & NodePort	20 min
4:35 – 4:45	Topic 9: Storage, Jobs & CronJobs	10 min
4:45 – 5:00	Lab 18: Kubernetes Storage — emptyDir, PV & PVC	15 min
5:00 – 5:15	Lab 19: Jobs & CronJobs — Batch and Scheduled Tasks	15 min
5:15 – 5:30	Course review, learner feedback & closing	15 min

Labs Covered

All 19 hands-on labs are delivered in KillerCoda browser terminals and mirror the slide deck and Learner Guide exactly:

Lab	Topic	Title
1	Docker commands	Docker Commands — Run & Manage Containers
2	Docker commands	Images & Inspecting Containers — pull, cp, inspect

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3	Docker Build / Dockerfile	Build the TaskBoard Image with a Dockerfile
4	Dockerfile best practices	Dockerfile Best Practices — .dockerignore & layer caching
5	CMD vs ENTRYPOINT	CMD vs ENTRYPOINT — the taskboard-cli tool
6	Volumes	Docker Storage — Named Volumes & Bind Mounts
7	Networking	Docker Networking — Custom Bridge, DNS & Port Mapping
8	Environment variables	Configuration with Environment Variables
9	Docker Hub	Sharing Images — Push to Docker Hub
10	Docker Compose	Docker Compose — Single Service
11	Docker Compose	Docker Compose — Multi-Service (Web + Redis)
12	Docker Compose	Docker Compose — Full-Stack (Web + PostgreSQL + Redis)
13	Pods	Kubernetes Pods — Imperative & Declarative
14	Namespaces	Kubernetes Namespaces — Environment Isolation
15	Deployments	Deployments — Scaling & Self-Healing
16	Rolling updates & rollbacks	Rolling Updates & Rollbacks
17	Services	Services — ClusterIP & NodePort
18	Storage	Kubernetes Storage — emptyDir, PV & PVC
19	Jobs & CronJobs	Jobs & CronJobs — Batch and Scheduled Tasks

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